

“We are seeing demand for both low- and high-end 32-bit MCUs”



Michel Buffa, microcontroller division general manager, STMicroelectronics, talks to Rahul Chopra from Electronics For You about the fundamental technological differences between 8-, 16- and 32-bit microcontrollers and ST's business strategy around these

■ MICHEL BUFFA
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What is your opinion on 16-bit microcontrollers (MCUs)? Are these facing an eventual demise?

If you look at ST's MCU product range, you will find that we have a big concentration of 8-bit and 32-bit MCUs, and we are not investing anymore in 16-bit MCUs. The reason why we believe that 16-bit microcontrollers (MCUs) will disappear, in fact, is linked to the layout of the device and the corresponding size occupied by the core only in the layout. Be it a 16-bit or 32-bit core, the contribution of the core to the die size of the device is in the range of 5 per cent to 7 per cent. Therefore there isn't much difference in terms of price between a 16-bit and a 32-bit MCU core. Basically, at the price of a 16-bit MCU, you can get a 32-bit MCU.

How do these 32-bit MCUs fare in terms of performance?

When you look at our range of 32-bit MCUs, there is a slight overlap between the 8-bit and 32-bit MCUs, which means we have a low-end 32-bit MCU that will be able to catch up with the 8-bit MCU. There is an overlap in terms of pricing but with the benefit of higher performances for the low-end 32-bit compared to 8-bit.

When we talk to design engineers, typically, we get a view that ST is a brand yet to be built in the MCU space. How are you tackling this challenge globally?

We were not big in terms of MCUs in the past. In 2009, we were No. 11 and then moved up to No. 8 in 2010. We

have been able to register a growth of 104 per cent. Essentially, our STM8 and STM32 product families have got a huge acceptance in the market and these are the major contributors to our growth.

Are there any specific applications for which you see a market for 32-bit MCUs in Asia?

We can have 32-bit MCUs everywhere. These can be used in industrial, point-of-sales, metering, medical, audio and a host of other applications.

Given the fact that mostly the low-end or mass volume manufacturing is happening in India, do you see a market now?

It is clear that for India, the market is more 8-bit centric. But there are many segments and applications in India where we are seeing more and more demand for both low-end and high-end 32-bit MCUs.

Essentially, your engagement strategy with respect to India was to pick up the leading technology stack—STM32—and get the community coming in. Is there any other engagement strategy besides it?

I think we were the first to choose cortex M as the core platform. For sure, we have to use some outsourced technology that is available with our competitors also. But the main differentiation comes at the ultra-low-power line of STM32,